

ML-Based 5G Traffic Generation for Practical Simulations Using Open Datasets

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The authors created a 5G traffic dataset by measuring various packet traffic and made it publicly available for anyone to use.

ABSTRACT

As artificial intelligence (AI) and machine learning (ML) technologies continue to advance, they are creating innovations in various industrial sectors and are increasingly influencing the development of wireless communication networks. Research advances in ML for wireless communications and networks depend essentially on the availability of datasets for testing results and attempting generalizations, but the availability of datasets obtained from real-world networks or experimental testbeds is currently very limited. To address this issue, we created a 5G traffic dataset by measuring various packet traffic and made it publicly available for anyone to use. The dataset is 328 hours long, making it difficult to handle directly and requiring preprocessing to be used for a specific purpose. To overcome these limitations, this study implements two types of traffic generators that can generate 5G traffic by improving ML models that have been verified and are widely used in academia. The traffic generator trained on the collected dataset has millions of parameters, but the total size of the model is no more than 118.12 MB, making it lightweight and easily deployable. To evaluate the difference between the generated traffic and the real traffic, the Jensen-Shannon divergence (JSD) and maximum mean discrepancy (MMD) metrics are observed, and it shows that the generated traffic is very similar to the real traffic. Our implementation and pretrained models are available on IEEE Code Ocean and GitHub.

INTRODUCTION

The successful integration of rapidly evolving modern machine learning (ML) models into over-the-air communication depends on the availability of datasets for testing results and attempting generalizations. Creating standard reference datasets for research purposes, from low-level physical layer measurements to upper-layer network analyses, is expensive for research institutions, and most datasets from mobile network operators are inaccessible. Although several real-world network datasets are publicly available [1, 2], applying them to various use cases is challenging because they were measured in specific situations and their volumes are insufficient to train state-of-the-art deep neural network models.

We have collected real 5G traffic and made the dataset publicly available for anyone to use. Our dataset covers a wide range of applications, including video streaming, videoconferencing, gaming, and the metaverse. However, raw datasets, including our published ones, require data preprocessing to suit their intended purpose and are inconvenient to handle due to their large size. ML models trained on datasets have numerous parameters, but if the input/output dimensions can be customized to fit the purpose, such models are lighter and more convenient than directly processing real traffic traces. Trained ML models that can generate different traffic patterns based on given conditions are useful for practical simulations.

Although ML models for predicting or classifying packet traffic have been extensively studied, research on ML models that generate a substantial amount of traffic based on actual datasets has been rarely conducted. Cheng [3] proposed and prototyped a generative adversarial network (GAN) model for generating realistic packet traffic such as ping, domain name system (DNS), and hypertext transfer protocol (HTTP). While the GAN has a generator structure that performs simple upsampling, its use for generating packets is novel compared to existing autoregressive models. Shahid *et al.* [4] proposed combining an autoencoder with a GAN to generate sequences of packet sizes that behave closely to real bidirectional Internet of things (IoT) flows. The autoencoder is trained to compress and reconstruct real sequences of packet sizes, and a GAN is then trained in the latent space to generate realistic sequences.

This article presents two ML models for 5G traffic generation. The first is the 5G traffic GAN (5GT-GAN) model, whose generators are trained to generate different types of traffic based on conditional inputs. Since the generator has a long short-term memory (LSTM) structure, it generates time series traffic with a probability distribution similar to that of the training dataset. The discriminator is designed to have a very large receptive field while maintaining the resolution and computational efficiency of the input by applying the dilated convolution technique to learn the characteristics of video traffic with long-term dependencies. The second model partially modifies the neural hierarchical interpolation for time series forecasting (N-HITS) approach [5],

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which achieved state-of-the-art performance in time series prediction, to be suitable for 5G traffic generation. To improve the forecasting output, a 1D-CNN layer and a gated recurrent unit (GRU) are added, and a postprocessing strategy is included to continuously generate a significant amount of traffic.

The main contributions of this article are summarized as follows:

- The traffic we collect covers a wide range of applications, from video streaming to the metaverse. Our dataset is available at [6].
- Two types of ML models that can generate various types of 5G traffic are developed based on well-known models. The dataset we collected is used to train the models. Our implementation and pretrained models are given on GitHub and IEEE Code Ocean [7].
- A traffic simulator equipped with the proposed traffic generation models is implemented to enable mixed simulation of various types of 5G traffic.

The remainder of this article is organized as follows. The next section describes the 5G traffic dataset. We then describe two ML models to replicate it, and present the evaluation setup, model training, and key results. Finally, we conclude this article.

5G DATASET COLLECTION

Representative applications that can collect 5G datasets directly from mobile terminals without using specialized equipment include G-NetTrack Pro and PCAPdroid. The former allows for monitoring and logging of the header and payload information of the medium access control (MAC) frame passing through the 5G air interface. The latter is an open-source network capture and monitoring tool that works without root privileges and analyzes connections made by applications installed on the user's mobile device. It can also dump mobile traffic to PCAP (also known as libpcap) and send it to the well-known Wireshark for further analysis.

We created a 5G dataset by measuring 5G traffic directly from a major mobile operator in South Korea. The model name of the mobile terminal used for traffic measurement is the Samsung Galaxy A90 5G, equipped with a Qualcomm Snapdragon X50 5G modem. We installed PCAPdroid, a packet sniffer software, on the terminal via Google Play. Traffic was measured sequentially per application on two stationary terminals (only one terminal is used for non-interactive services) with no background traffic. The dataset includes a diverse number of traffic types and is shown in Table 1. The collected dataset includes resource-intensive video traffic that has the greatest impact on 5G network planning and provisioning. We did not mix background traffic to measure the unique characteristics of each type of traffic.

The video streaming dataset contains data directly measured while watching Netflix and Amazon Prime Video, representative over-the-top (OTT) services, on mobile devices. The live streaming dataset is measured while watching YouTube Live and South Korea's famous live broadcasts (Naver NOW and Afreeca TV). Video conferencing data are measured by conducting live meetings on the popular Zoom, MS Teams, and Google Meet platforms. Two types of metaverse traffic are acquired: Zepeto and

Type	Application	Protocol	Duration and size
Live Streaming	YouTube Live	GQUIC	20h 19m 38s File size: 0.73GB
	AfreecaTV	TCP	20h 14m 00s File size: 4.06GB
	Naver NOW	TCP	33h 50m 34s File size: 12.48GB
Stored Streaming	YouTube	QUIC	22h 59m 51s File size: 1.12GB
	Netflix	TCP	24h 43m 02s File size: 0.74GB
	Amazon Prime Video	TCP	32h 39m 10s File size: 1.54GB
Video Conferencing	Zoom	UDP	26h 12m 53s File size: 3.36GB
	MS Teams	UDP	28h 17m 27s File size: 3.71GB
	Google Meet	UDP	24h 01m 40s File size: 4.41GB
Metaverse	Zepeto	TCP	15h 28m 36s File size: 0.16GB
	Roblox	RakNet	25h 04m 11s File size: 0.11GB
Online Game	Teamfight Tactics	UDP	13h 46m 53s File size: 0.24GB
	Battleground	UDP	16h 02m 57s File size: 0.38GB
Game Streaming	GeForce Now	UDP	12h 26m 21s File size: 7.05GB
	KT GameBox	UDP	12h 23m 26s File size: 4.36GB

TABLE1. Dataset [6] configuration.

Roblox. Zepeto traffic is collected while staying in "Camping" for 15 hours. Roblox traffic is collected by playing "Collect All Pets" for 25 hours using the auto-clicker. We collect two types of mobile network gaming traffic. The first is cloud gaming, an online game setup that runs video games on remote servers and streams them directly to the user's device. The second is a typical mobile game connected to the Internet.

The dataset was collected from May to October 2022, has a total length of 328 hours, and is provided in CSV file format. The dataset is a timestamp-mapped time-series dataset with packet header information, and further traffic analysis by application is possible because it includes source and destination addresses. To make it more usable as a traffic source model, we describe how to use it as a training dataset for the traffic simulator platform's source generator in the next section.

TRAFFIC GENERATION MODELS

The 5G traffic dataset measured by PCAPdroid can be used as a training dataset for various ML models. In this study, we implemented two ML models that can generate traffic similar to the data in the collected dataset based on input conditions. Although the pretrained models have sizes of around tens of megabytes, they have the advantage of continuously generating vari-

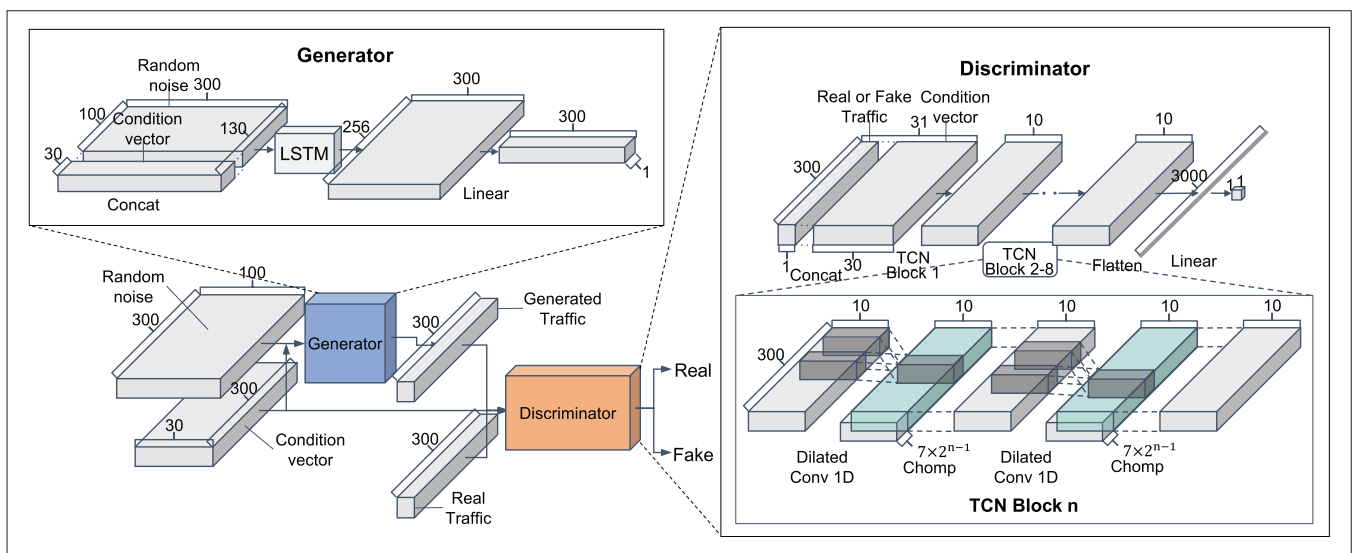


FIGURE 1. 5GT-GAN architecture.

ous types of 5G traffic similar to real traces. The two traffic generation models are included in the traffic simulator developed in Python to increase user convenience.

5GT-GAN

5GT-GAN is an improved model that can generate various types of 5G traffic from a single generator by applying the recurrent conditional GAN (RCGAN) [8] structure as a modification of the previously proposed model [9]. As shown in Fig. 1, 5GT-GAN comprises two neural networks: a generator and a discriminator. Similar to a regular GAN, these two neural networks compete with each other in a two-player minimax game, where one player's gain is another player's loss. The generator's output is directly connected to the discriminator's input through a hyperbolic tangent activation function that normalizes the training data to values in the range $[-1, 1]$. When the training process begins, the generator produces random data according to the given conditions, and the discriminator learns that these data are not 5G traffic data. However, if the generator is effectively trained, it can produce new data instances that resemble the ground-truth data, and the discriminator can no longer differentiate between real and fake data.

The discriminator comprises eight temporal convolutional network (TCN) blocks. Each TCN block concatenates a condition matrix with a size of 300×30 and real or generated traffic with a size of 300×1 , and alternately uses it as inputs to the next layer. It has a structure in which a 1D-CNN and a Chomp layer are repeated twice. By stacking TCN blocks with dilated convolution, the network can have very large receptive fields with only eight layers, and maintain the input resolution and computational efficiency across the network. In this article, the dilation rate is doubled for every layer up to a limit of 128. Training is performed until the discriminator output value approaches 0.5, making it difficult to distinguish between real (1) and fake (0) data. The discriminator's loss function is binary cross-entropy, and the total number of learnable parameters is 16,841.

The generator creates traffic similar to real traffic. A latent space of size 130×300 is used as input to an LSTM with 256 hidden cells. To generate traffic of size 1×300 , we serialize the output of the LSTM into a vector and use it as the input to the fully connected layer. The latent space refers to an abstract multidimensional space containing feature values that cannot be directly interpreted, but it encodes internal representations of externally observed data. In this article, the latent space is a hypersphere created by concatenating a 100×300 matrix with each variable derived from a standard normal distribution and a 30×300 condition matrix. We use the feature matching loss [10], a loss function that learns by comparing the feature of the real trace extracted from each temporal block of the discriminator with the feature of the generated traffic, so that the generator can clearly reflect the features of real traffic. The total number of learnable parameters for the generator is 385,281.

N-HITS FOR 5G TRAFFIC

The latest trend in time series data forecasting is to prevent error accumulation and improve inference speeds by forecasting multiple steps at once, rather than performing recurrent forecasting. Two state-of-the-art approaches are observed in the field of multistep forecasting. The first is to explicitly model the interactions between inputs and outputs using a transformer-based attention mechanism [11, 12], and the second is to achieve improved computational efficiency by implicitly modeling the input-output relationships based on fully connected stacks [5, 13]. Both approaches are evolving to better capture the long-range dependencies of time series by improving the capacity of the associated models. However, long-horizon forecasting is a difficult task because forecast values are volatile, and the memory and computational complexity increase as the scales of the attention mechanism and fully connected layer increase. If sufficiently long multistep predictions can be made accurately and multistep prediction values can be continuously generated, the prediction values can be used as a traffic source for a traffic simulator.

N-HITS [5] is an improved version of the

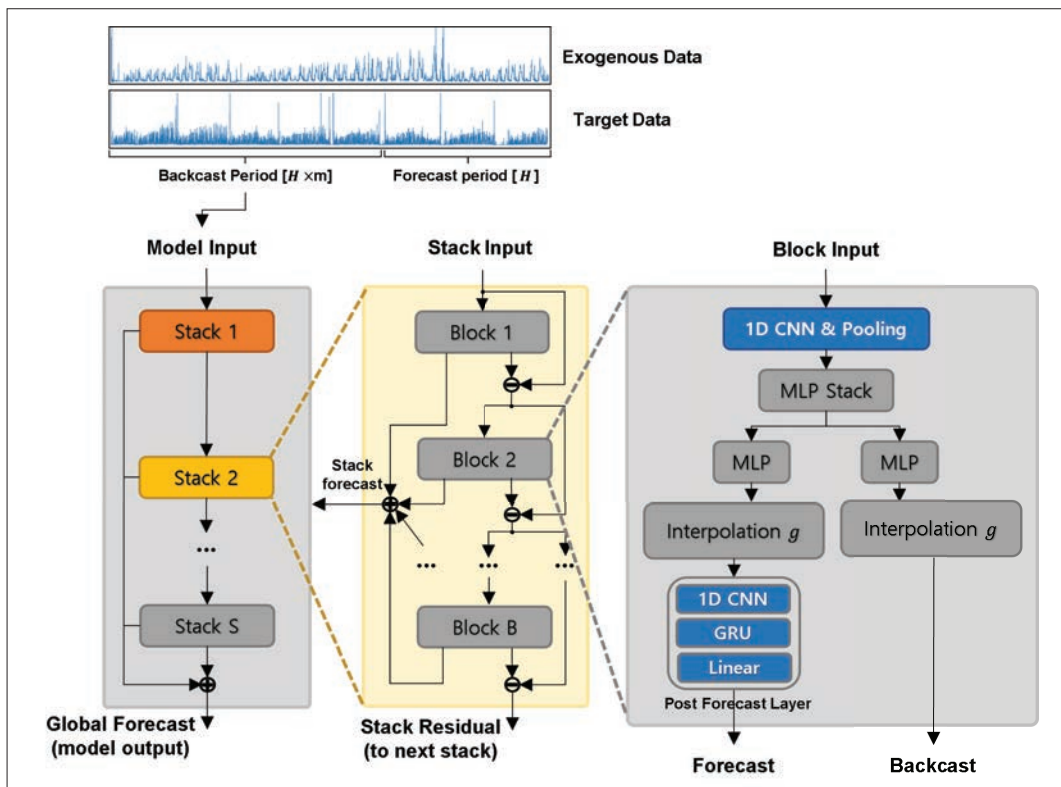


FIGURE 2. N-HITS-5G model architecture. Exogenous variable [14] is an input that affects the target variable. When training downlink traffic, uplink traffic of the same time is used as an exogenous variable. Blocks added in this study are shown in blue.

N-BEATS [13] model in several important ways, making it more accurate and computationally efficient, especially in the context of long-horizon forecasting. N-HITS uses multi-rate sampling for the input signals and multiscale synthesis for the forecasts, resulting in a hierarchical forecast construction process. This greatly reduces the computational requirements and provides improved forecasting accuracy. It also provides interpretable time series decomposition. N-HITS is a state-of-the-art model that outperforms Autoformer [11] and Informer [12] in time series data forecasting.

In this work, we extend the existing N-HITS architecture, as shown in Fig. 2, by adding two more layers to enhance its input decomposition process via multi-rate data sampling and by modifying its output synthesizer to synthesize packet traffic. We also replace the forecast output layer of N-HITS with ours so that the system can generate continuous 5G traffic. N-HITS for 5G traffic (N-HITS-5G) decomposes 5G traffic traces to capture time series features such as trends and seasonality without using a statistical approach. The upper stack (Stack 1) is designed to predict trends, which is a low-frequency component, the middle stack (Stack 2) is designed to predict seasonality, and the lower stacks are designed to predict the residuals, which are low-frequency components. We add a 1D-CNN block suitable for learning the characteristics and patterns of 5G traffic to improve the output of each stack in the proposed model. In addition, while the existing model has the same structure for the forecast output and backcast output, the proposed model combines the forecast output with a GRU unit to provide improved forecasting performance.

The input data are scaled to values between $[-1, 1]$ and used for training. The number of learnable parameters of the N-HITS-5G model ranges from a minimum of 10.3 M to a maximum of 29.5 M, and in terms of size, it ranges from 41.29 MB to 118.12 MB. Depending on the type of traffic, different horizontal lengths and hyperparameter tuning are performed.

IMPLEMENTATION AND EXPERIMENTAL RESULTS

This section first describes the details of training and evaluation setups. We then describe the key results, limitations, and future research directions.

EVALUATION SETUP AND MODEL TRAINING

For the implementation of the proposed 5GT-GAN and N-HITS-5G models, we used the PyTorch Lightning library of the PyTorch framework on Docker running on Ubuntu 18.04 LTS. Specifically, we used one 32-GB Nvidia Tesla V100 GPU for training 5GT-GAN and another 80-GB Nvidia DGX A100 GPU for training N-HITS-5G. For both models, we split the dataset into 70 percent for training, 10 percent for validation, and 20 percent for testing.

To train 5GT-GAN, we used the following settings and hyperparameters: the number of temporal blocks is 8, the convolution kernel size is 8, the stride is 1, and the dilation rate consists of $\{1, 2, 4, 8, 16, 32, 64, 128\}$. We set the learning rate to 0.0002, used the Adam optimizer, set the batch size to 64, and trained the model for 300 epochs. In our computing environment, the training time per epoch was approximately 50 m.

We explored the hyperparameters of N-HITS-5G using *hyperopt*, a Bayesian hyperparameter optimization library. We set the learning rate to 0.001 and the batch size to 20. The model was

trained for up to 1000 epochs, and the training process was designed to stop early if the loss did not decrease by more than 0.0001 over 10 epochs. Most of the settings were kept the same as those in the original N-HiTS, and the param-

eters added as the model was modified are summarized in Table 2.

KEY RESULTS

Figure 3 depicts the traffic volume patterns generated over time by the 5GT-GAN and N-HiTS-5G models. When 5GT-GAN generates traffic in the same computing environment as the learning environment, it takes only 450 ms to generate 15 types of 5-minute traffic for both uplink and downlink, which means that 150 minutes of traffic can be generated in 450 ms. In contrast, it takes approximately 30 s to generate 5 minutes of traffic using the N-HiTS-5G model. N-HiTS-5G has separate pretrained models for each type of traffic, and each model needs about 10 s to load and 20 s to infer. If there is no memory limitation, loading all pretrained ML models in advance when the simulator starts can reduce traffic generation time. It is also worth considering changing the GRU module used in the post-forecast layer to a non-autoregressive structure with fewer parameters to improve inference speed. Fig. 3 shows excerpts of 2 types of traffic generated by 5GT-GAN and N-HiTS-5G.

Hyperparameters	Considered Values
Input size multiplier (m)	$m \in \{3, 5, 10\}$
Pooling kernel size (k_i)	$[k_1, k_2, k_3] \in \{[8, 4, 1], [16, 8, 1], [1, 2, 4], [1, 2, 8]\}$
Number of Stack's coefficients (r_i^{-1})	$[r_1^{-1}, r_2^{-1}, r_3^{-1}] \in \{[168, 24, 1], [24, 12, 1], [180, 60, 1], [40, 20, 1], [64, 8, 1], [8, 4, 2], [24, 8, 2]\}$
Interpolation strategy $g(\tau, \theta)$	$g(\tau, \theta) \in \{\text{linear}, \text{cubic}\}$
1D-CNN Pooling layer padding size (p_i)	$[p_1, p_2, p_3] \in \{[3, 1, 0], [7, 3, 0], [0, 0, 1], [0, 0, 3]\}$
1D-CNN Post forecast layer's kernel size (k_f)	$k_f \in \{[5, 24, 48, 96, 168, 360]\}$
GRU Post forecast layer's hidden size (h)	$h \in \{8, 16, 32\}$
Forecast Horizon (H)	$H \in \{[10, 48, 96, 192, 336, 720]\}$

TABLE 2. Evaluation setup for N-HiTS-5G.

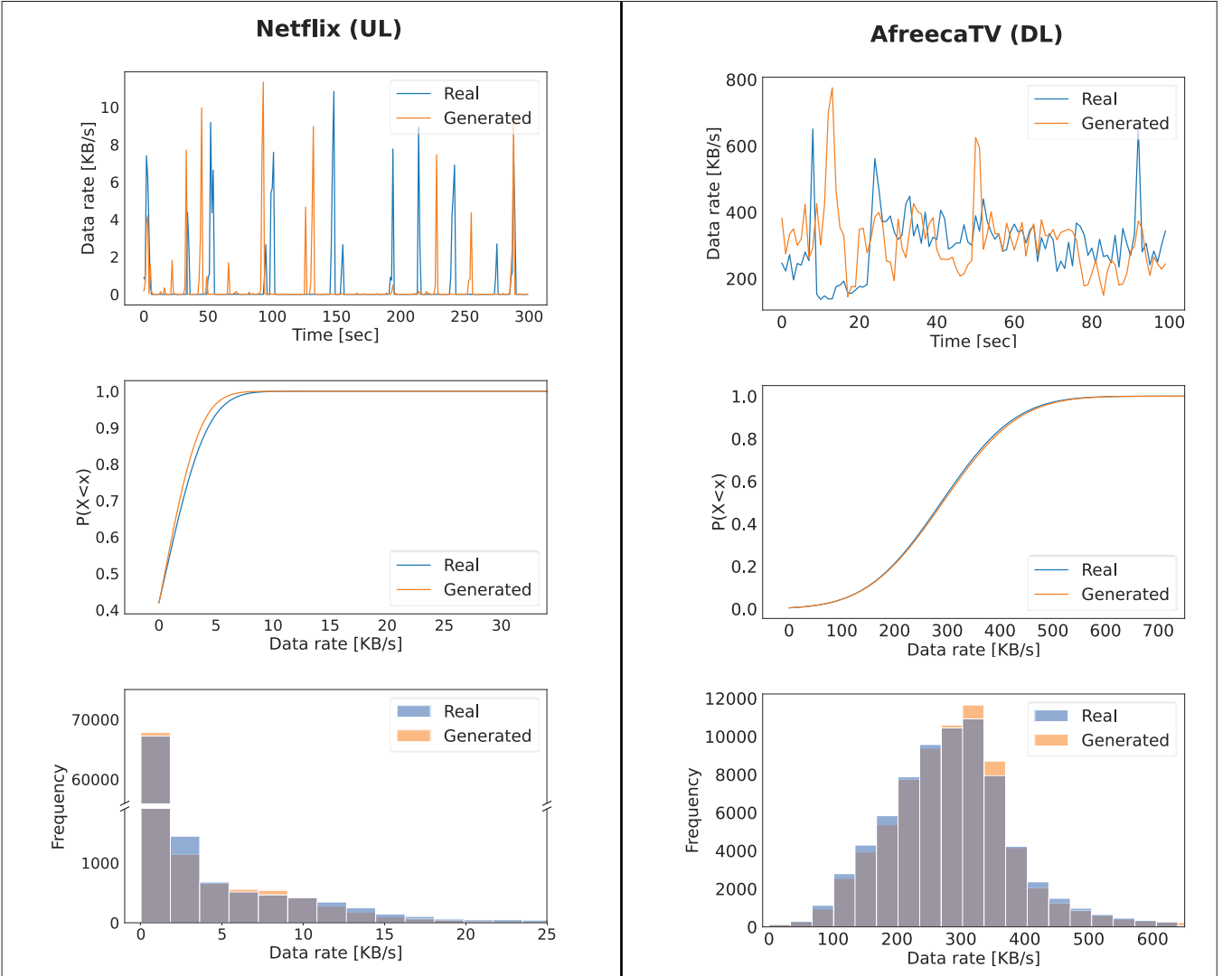


FIGURE 3. It shows the traffic patterns generated by two different models, 5GT-GAN and N-HiTS-5G, for two different types of traffic: Netflix uplink traffic and AfreecaTV downlink traffic. In the upper left corner, the figure displays the Netflix uplink traffic pattern generated by the 5GT-GAN model. The center and bottom of the figure show the corresponding CDF and histogram, respectively. On the top right, the figure displays the AfreecaTV downlink traffic pattern generated by the N-HiTS-5G model. Again, the center and bottom of the figure show the corresponding CDF and histogram, respectively.

In web-based video streaming (e.g., Netflix and Amazon Prime Video), downlink traffic patterns are mostly affected by the browser's request cycles, which exhibit irregular characteristics depending on the client software configuration, network congestion, and load balancing algorithm. As shown in Fig. 3, request intervals for Netflix are highly irregular, ranging from a few tens of milliseconds up to 74 seconds. Converting this to the number of data samples results in a long-term dependent process with autocorrelation lags up to 3,700 samples. Therefore, stochastic models such as the autoregressive integrated moving average (ARIMA) model with only a few dozen parameters are insufficient to represent this type of time series.

To evaluate the proposed traffic generation models' performance more accurately, we use the Jensen-Shannon divergence (JSD) and maximum mean discrepancy (MMD) metrics. JSD measures the similarity between two distributions P and Q , where P is the probability distribution generated by the proposed model, and Q is the probability distribution of the real dataset. The JSD measure is a symmetrized and smoothed version of the Kullback-Leibler (KL) divergence. MMD is a kernel-based statistical test that measures the distance between the two distributions P and Q .

Table 3 shows the performance evaluation results obtained for 15 types of 5G traffic generated by the 5GT-GAN and N-HITS-5G. As shown in Table 3, JSD values between 0.00001 and 0.01316 and MMD values between 0.00002 and 0.76559 are observed. It can be seen that the high-capacity N-HITS-5G model generates 5G traffic more accurately in most traffic types. This indicates that the distributions of generated traffic and ground truth are very similar except for the online game traffic of 5GT-GAN, and it can be seen that the ML models generate traffic that is similar to the real traffic traces.

We implemented a simple traffic simulator equipped with the two proposed ML models, allowing users to create a scenario in which 15 types of traffic are mixed and observe the generated traffic as shown in Fig. 3. When the simulation is finished, the raw data are stored as a file in CSV format, and various statistical information such as the average and standard deviation of the bit rate are summarized. In addition, packet size distribution, packet inter-arrival time distribution, and bit rate distribution of generated traffic can be viewed as cumulative distribution function (CDF) graphs, and histograms and box plots are also provided.

Our ML models do not provide high-level control over traffic characteristics affected by various radio channel conditions, and do not model traffic patterns generated by moving terminals, thus limiting their practical application. Motivated by style transfer literature [15], we are gradually upgrading the generator architecture of 5GT-GAN to control the traffic synthesis process. The new architecture allows for unsupervised separation of high-level attributes from stochastic variation in the generated traffic and enables mixing and interpolation operations to adjust the output style. Since it is practically impossible to collect a large amount of 5G traffic containing various travel speeds, research is also underway to fine-tune the pre-trained N-HITS-5G by additionally learning a small amount of dataset measured in a mobile environ-

Type	Application	JSD	MMD
Live Streaming	YouTube Live	0.00193 0.00249	0.01029 0.01445
	AfreecaTV	0.00076 0.00010	0.00037 0.00002
	Naver NOW	0.00343 0.00015	0.05447 0.00287
Stored Streaming	YouTube	0.01057 0.00005	0.00115 0.00237
	Netflix	0.00219 0.00051	0.00028 0.00092
	Amazon Prime Video	0.00036 0.00021	0.00036 0.00680
Video Conferencing	Zoom	0.01316 0.00303	0.04183 0.00005
	MS Teams	0.00403 0.00730	0.03248 0.00005
	Google Meet	0.00130 0.00049	0.00050 0.00002
Metaverse	Zepeto	0.00006 0.00036	0.05393 0.00464
	Roblox	0.00171 0.00161	0.00673 0.00013
Online Game	Teamfight Tactics (TFT)	0.00448 0.00001	0.31550 0.00046
	Battleground	0.00922 0.00136	0.76559 0.07324
Game Streaming	GeForce Now	0.00972 0.00042	0.00175 0.00040
	KT GameBox	0.00947 0.00274	0.00592 0.00005

TABLE 3. Performance of the 5GT-GAN and N-HITS-5G. Lower scores are better. The upper number in each cell is the 5GT-GAN score, and the lower number is the N-HITS-5G score.

ment. In this case, a separate ML model is created depending on the dataset used for fine-tuning.

CONCLUSION

We collected and published a real 5G traffic dataset containing packet header information for diverse applications. However, due to the high volume of the collected data, it is inconvenient to use in practice. To address this issue, we have developed two types of traffic generators for 5G traffic by improving widely used and validated ML models in the literature. These traffic generators are trained on our collected dataset, have millions of parameters, and are smaller than 118.12 MB in size. They can generate 75 minutes of traffic in less than 1 second, making them ideal for simulation or testbed load generation. The two ML models proposed in this study can replace traditional stochastic traffic source models within network simulators and are particularly well-suited for volume-based traffic simulations. They have the potential to be utilized in various 5G network planning and provisioning tools, including a private 5G spectrum requirement calculator [9], which estimates the amount of spectrum needed for a specific network deployment.

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